

Break-even Analysis [B-E]

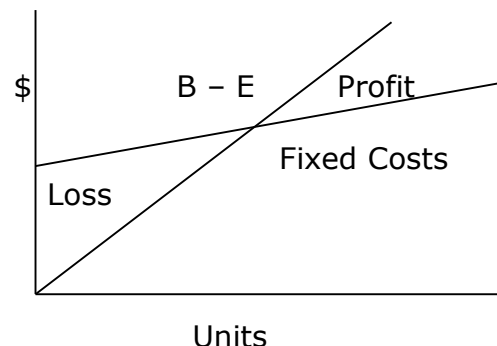
1. Break – Even Analysis is the analysis of Cost versus Income
2. Cost has 2 components
 - a) Cost
 - i) Fixed Cost including Capital Cost, Cost of land, Taxes (GST),
 - ii) Variable Cost is the Production Cost, manpower, material machinery, technology, O&M, Utility Bills, Salaries, Rent etc.
3. Students have learnt Factors of Production: Land, Labour, Capital. Students have also studied Laws of Return, Law of Increasing Return, Law of Declining or Decreasing Return and Law of Constant, O&M, Utility Bills, Staff Salaries, rent.
 - b) Income or return on investment also called Receipts or Revenue
4. In B-E Analysis a point is reached when Cost = Income called Break – Even Point (BEP)
5. Break Even Point is attained when there is:

No profit

No loss

or No profit, no loss
6. By suitable combination of Fixed Cost and Variable Cost and marketing profit can be earned.
7. If income is higher than cost there is profit.
8. If income is lower than cost there is loss.
9. Equation

$$\begin{aligned}
 \text{TR} &= \text{TC} \dots\dots\dots(1) \\
 P \times X &= \text{TFC} + V \times X \dots\dots\dots(2) \\
 P \times X - V \times X &= \text{TFC} \dots\dots\dots(3) \\
 (P - V) \times X &= \text{TFC} \dots\dots\dots(4) \\
 X &= \frac{\text{TFC}}{P - V} \dots\dots\dots(5)
 \end{aligned}$$



Where

TR = Total Revenue

TC = Total Cost

TFC = Total Fixed Costs

P = Unit Sale Price

V = Unit Variable Cost

X = No. of Units

Problem

A manufacturing industry manufactures water Pressure Controlling Valves. Given values are as under:

Price of Valve = \$ 15/valve

Cost Variable valve = \$ 7/valve

Capital or Fixed of Cost Plant = \$ 9000

Calculate: a) B – E Point or BEP

b) How Manufacturer can earns profit

Solution

p = \$ 15

v = \$ 7

FC = \$ 9000

Substitute Values in Equation (4)

$$x = \frac{TFC}{p - v}$$

$$x = \frac{9000}{(15 - 7)}$$

$$x = \frac{9000}{8}$$

$$x = 1,125 \text{ units}$$

$$BEP = 15 \times 1,125$$

$$BEP = \$16,875$$

$$\text{Now } \$ 16,875/15 = 112.4 \text{ Valves (BEP)}$$

The manufacturer must sale at least 112.4 or 112 valves at \$ 15/valve to earn BEP.

The manufacture can increase sale of more than 112 valves to earn profit.

Conclusion:

a) Manufacturer should sale 112 Water Pressure Controlling Valve to attain BEP

b) Manufacture should sale more than 112 Water Pressure Controlling Valves to earn profit.

Problem

Suppose Management of a firm has received 2 alternate electric motors that provide 100 hp output (1 hp = 0.74 kw). The given values are as follows:

Detail	Motor A	Motor B
Initial Cost (P)	\$ 12,500	\$ 16,000
Life (n)	10 years	10 years
O&M Maintenance	\$ 500 / year	\$ 250 / year
Efficiency	74%	92%
Interest rate (i)	15%	15%
Taxes	1.5% / year	1.5% / year
Electricity Consumption	74.6 kw	74.6 kw
Full Load Hours of A & B	1100 hours	1100 hours
Electricity Charges	\$ 0.05 kwh	\$ 0.05 kwh

Calculate following values:

- Equivalent Annual Uniform Cost (EAUC) of A and B
- Electricity Consumption Cost of both alternates
- Taxes of both Alternates
- O & M of both Alternates
- Assume "x" as equivalent hours of operation where costs of Motor A and Motor B coincide
- Find value of Motor A and Motor B if Electric Motors are operated for 1100 hours /year
- State whether Motor A is economical or Motor B is economical

Solution

Motor A

$$EUAC_A$$

Or

$$EUAC_A = P \left[\frac{i(1+i)^n}{(1+i)^n - 1} \right]$$

$$EUAC_A = 12500 \left[\frac{0.15(1+0.15)^{10}}{1+0.15)^{10} - 1} \right]$$

$$EUAC_A = 12500 \left[\frac{0.604}{3.04} \right]$$

$$EUAC_A = 12500 \left[0.1993 \right]$$

$$EUAC_A = \$ 2490 \dots\dots\dots (a)$$

b) Electricity Charges

$$\frac{74.6 \times 0.05}{0.74} = \frac{3.73}{0.74} = \$ 5.04 \dots\dots\dots(b)$$

c) Taxes

Taxes : 1.5 %
Cost : \$ 12500

Therefore:

$$\begin{aligned} \text{Taxes} &= \$ 12500 (0.015) \\ \text{Taxes} &= \$ 187 \dots\dots\dots(c) \end{aligned}$$

d) O & M = \$ 500 / year \dots\dots\dots(d)

Add Motor A = Annual Cost + Electricity Charges + Taxes + O&M.

Assume x hours of operation against electricity charges to balance equation.

i.e.

$$a_A + b_A + c_A + d_A$$

$$\text{Motor A} = 2490 + 5.04x + 500 + 187$$

Values of Motor A

$$\begin{aligned} a_A + b_A + c_A + d_A &= 2490 + 5.04x + 500 + 187 \\ &= \$ 3177 + 5.04x \end{aligned}$$

Motor B

a) Annual Equivalent Cost

$$EUAC_B = 16000 \left[\frac{0.15 (1.15)^{10}}{(1.15)^{10} - 1} \right]$$

$$EUAC_B = \$ 3188.033 \dots\dots\dots (a)$$

b) **Electrical Cost**

$$\text{Electrical Cost} = \frac{74.6 \times 0.05}{0.92}$$

$$\text{Electrical Cost} = \$ 4.05 \dots\dots\dots (b)$$

c) **Taxes**

Taxes = 1.5 %
Cost = \$ 16000
Taxes = (16000 x 0.015)

$$\text{Taxes} = \$ 240 \dots\dots\dots(c)$$

d) O&M = \$ 250 \dots\dots\dots(d)

Values of Motor B

$$\text{Motor B} = a_B + b_B + c_B + c_B + d_B$$

Put "x" hrs of operation against electricity charges

$$a_B + b_B + c_B + c_B + d_B = 3188 + 4.05x + 240 + 250$$

Equate Project A = Project B

$$3177 + 5.04x = 3678 + 4.05x$$

$$5.04x - 4.05x = 3678 - 3177$$

x = 506.06 Hours when cost of Projects A and Project B are equal

Extrapolation

$$\text{EUAC}_A = \$ 2490 \text{ of } 506 \text{ Hours} = \$ 2490$$

$$\text{EUAC}_B = \$ 3188 \text{ of } 506 \text{ Hours} = \$ 3188$$

(f) EUAC_A of 1100 Hours

$$\frac{1100}{506} = 2.17 \text{ say } 2.2$$

$$\text{Motor A } 2490 \times 2.2 = \$ 5478$$

$$\text{Motor B } 3188 \times 2.2 = \$ 7013$$

For 1100 hrs operation Motor A is economical